ALICE L WALTON SCHOOL OF MEDICINE

# Cholinesterase inhibitors and reactivators

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Adapted from: Cholinomimetic Agents

Listen to this Brick

0:00 / 23:29 1x

**Learning Objectives (6)** Versions

***After completing this brick, you will be able to:***

1. Compare and contrast the mechanism of action of direct- and indirect- acting AChE inhibitors.
2. Describe the pharmacokinetic properties of various direct- and indirect- acting AChE inhibitors.
3. Recognize th ergic toxidromes and link them to cholinergic receptor activation.
4. Explain how cholinesterase reactivators (oximes) reverse organophosphate poisoning.
5. Provide the rationale for the use of atropine as an antidote for cholinergic poisoning.
6. Discuss the implications of organophosphate use in chemical warfare.

**CASE CONNECTION**

JK, a 35-year-old farm worker, has been spraying pesticides at his farm. Within an hour, JK was experiencing breathlessness and fatigue. He called his co-workers and told them that he was not feeling well. He also told them that he was very sweaty and had been to the bathroom 3 times in the last hour with diarrhea. His eyes were very watery and he was constantly wiping them. He told his co-worker that he was not able to see clearly now. His coworker got worried and drove JK to the nearest hospital (5 minutes away). When he reached the hospital, JK's heart rate was 45 beats per minute, respiratory rate was 28 breaths per minute, and body temperature was 37.2°C. He was in considerable respiratory distress and had to be given oxygen immediately. He stated that he was not able to see clearly. Examination showed that he has pinpoint pupils. You are a medical student in the ER with the attending physician. By now, JK, salivating continuously, has urinated and defecated on himself. JK was now showing muscle twitching and weakness. When the attending asked his coworker, he said that they were spraying the crops with parathion. The diagnosis of organophosphate poisoning was confirmed based on the clinica al student on rotation, while you are reviewing JK's chart you hear the physician order atropine and 2-PAM for the patient.

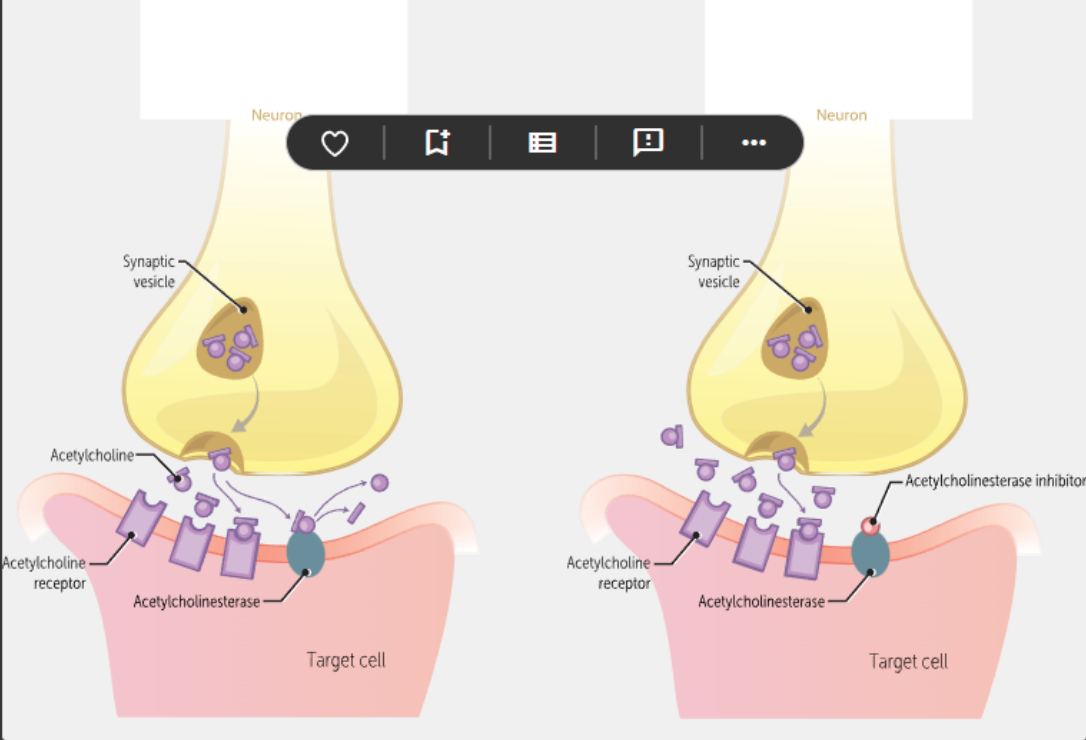
What confirms the physician's diagnosis of organophosphate poisoning? What could be the rationale behind ordering atropine and 2-PAM for JK? Consider your answer as you read, and we'll revisit JK at the end of the brick.

GO TO CONCLUSION ↓

## What Are Cholinomimetic Agents? (LO#1)

We have discussed cholinomimetics (or cholinergics) in the previous brick. A cholinomimetic is a substance that mimics (-mimetic) acetylcholine (cholino-). Acetylcholine is the primary distal neurotransmitter of the parasympathetic nervous system (PNS), activating the muscarinic receptors. There are two types of cholinomimetics.

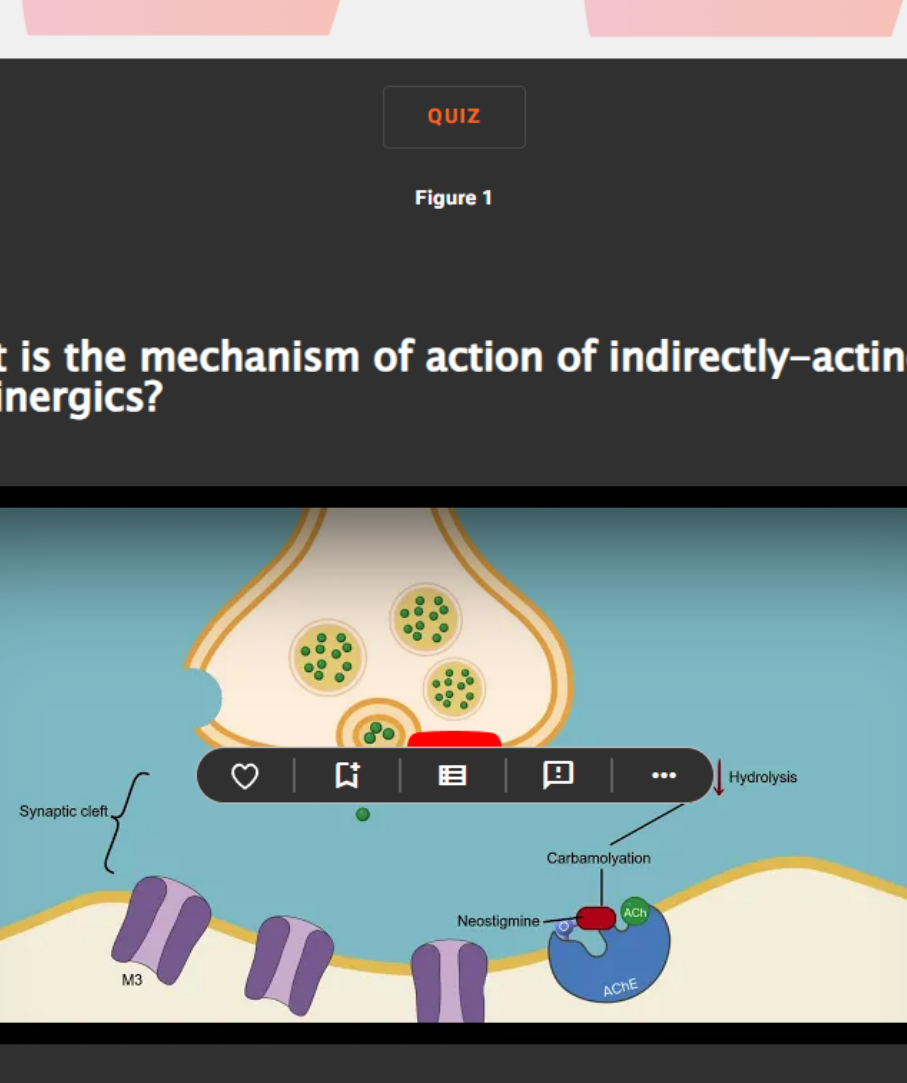
- **Direct-acting** cholinomimetic agents bind to acetylcholine receptors and cause activation of the PNS (eg. Pilocarpine).
- **Indirect-acting** cholinomimetic agents are also called **cholinesterase. inhibitors**, since they inhibit the breakdown of acetylcholine by the enzyme cholinesterase. Drugs in this class include neostigmine, organophosphates, etc. This leads to increased acetylcholine in the synaptic cleft, causing activation of the PNS (Figure 1).



QUIZ

Figure 1

## What is the mechanism of action of indirectly-acting cholinergics?



### What Are Common Indirect-Acting Reversible Cholinomimetic Agents (or cholinergics) and Their Clinical Uses?

Commonly used reversible indirect-acting cholinergics include:

- **Neostigmine** is used to manage glaucoma by reducing resistance to aqueous humor outflow. This is very similar to the mechanism of carbachol and pilocarpine discussed in section that covers the direct-acting agents.
- **Physostigmine** is used in the treatment of atropine overdose. Atropine is mostly used to increase the heart rate in arrhythmias by competitively inhibiting acetylcholine. Physostigmine will replenish the acetylcholine in the synapse, preventing anticholinergic toxicity. It is used to treat anticholinergic toxicity because of its ability to cross the blood-brain barrier.
- **Edrophonium** and **pyridostigmine:** These drugs may be used in myasthenia gravis (MG) in which acetylcholine stimulation is lacking in end organs. Edrophonium, a short-acting drug (5-15 minutes), briefly improves MG symptoms, confirming the diagnosis. Pyridostigmine has a longer duration and can be used for treatment.
- **Donepezil** and **rivastigmine** are used to treat Alzheimer disease and Parkinson disease. How they work in these disorders is not well understood, but i sterases. The drugs do not change the course or prognosis of the dementia but may slightly improve cognition for short periods.

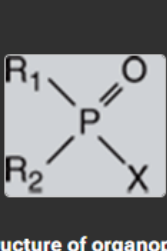
**CLINICAL CORRELATION**

Myasthenia gravis (MG) is an autoimmune condition that is characterized by presence of autoantibodies against acetylcholine receptors at the neuromuscular junction. MG leads to skeletal muscle weakness, including drooping eyelids, double vision, and difficulty in swallowing and talking. In advanced cases, MG may cause respiratory muscle dysfunction, leading to fatal pneumonia and respiratory failure.

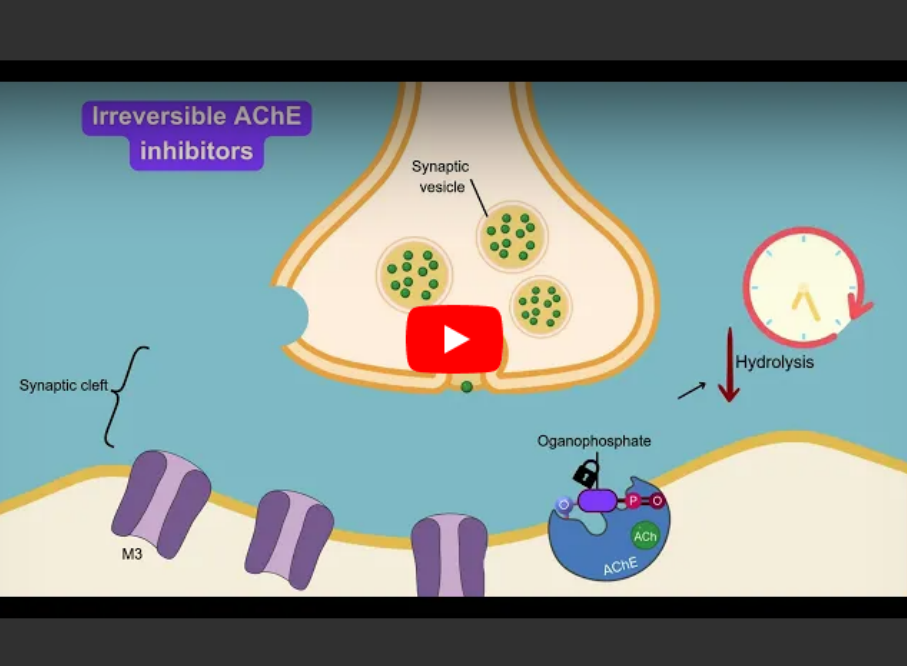
What is the primary use of neostigmine?

### With irreversible indirect-cholinesterase inhibitors

Organophosphorus inhibitors are tetrahedral in configuration, a configuration that resembles the transition state formed in carboxyl ester hydrolysis.



General structure of organophosphates



After the initial binding-hydrolysis step, the phosphorylated enzyme complex may undergo a process called aging. This process apparently involves the breaking of one of th nhibitor and further strengthens the phosphorus-enzyme bond. The rate of aging varies with the particular organophosphate compound.

What is the process called that strengthens the phosphorus-enzyme bond in organophosphate poisoning?

### What Are Common Indirect-Acting Irreversible Cholinomimetic Agents and Their Uses?

Most common use of irreversible cholinergic agents include insecticides such as parathion and malathion. Other examples include sarin, soman, VX etc. These agents are used in chemical warfare and can be extremely dangerous upon exposure.

## What are the ADME associated with indirectly acting cholinergics? (LO#2)

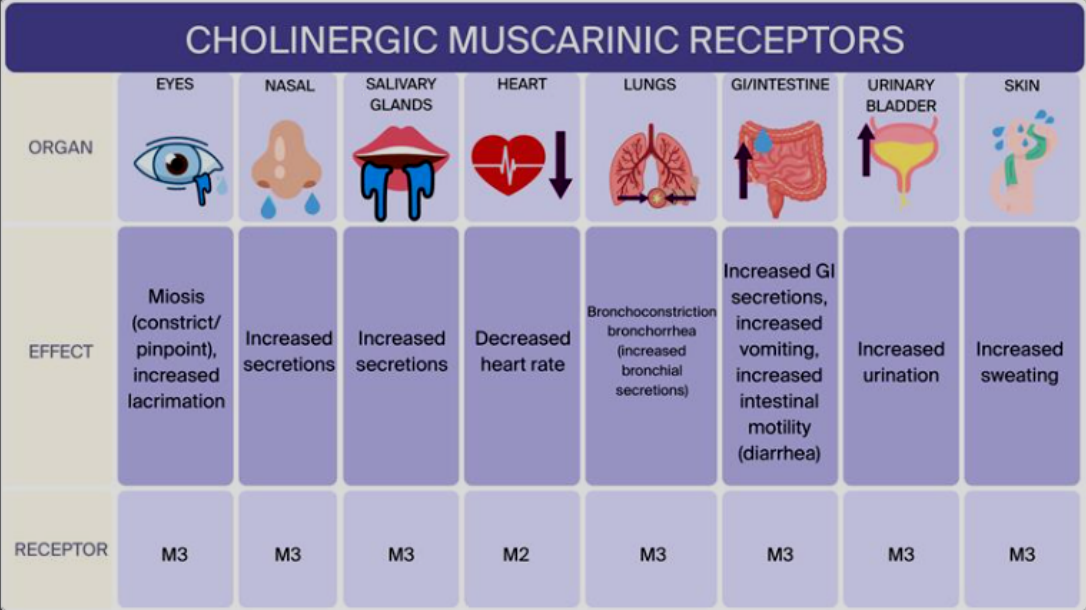
- Neostigmine and cannot cross the blood brain barrier. Physiostigmine is a 3° amine **which crosses** the BBB.
- The organophosphates are well absorbed from the skin, lungs, gut, etc. In general, they are **very lipid-soluble and have high $V_d$**. This makes them effective insecticides but also very dangerous to humans.

**CLINICAL REASONING**

**Physostigmine as antidote:** Physostigmine, is used specifically for anticholinergic poisoning. It is a reversible cholinesterase inhibitor that can cross the blood-brain barrier, making it effective in treating both peripheral and central symptoms of anticholinergic toxicity. By inhibiting acetylcholinesterase, physostigmine increases acetylcholine levels, which helps to reverse the effects of anticholinergic agents such as delirium, hallucinations, and agitation. Since it can cross the blood-brain barrier, it is often used in cases of severe anticholinergic poisoning where CNS symptoms are prominent.

## What Happens When Muscarinic Receptors Are Activated? (LO#3)

We have discussed the effects of activating cholinergic muscarinic receptors in the cholinergic and anticholinergics (1 of 2) brick. Recall that $M_1$-$M_3$ are the most important clinically. The receptors are found in many target organs, so the effects when they are activated by acetylcholine are extensive. The figure below describes the effect on these receptors.



CHOLINERGIC MUSCARINIC RECEPTORS

| ORGAN | EYES | NASAL | SALIVARY GLANDS | HEART | LUNGS | GI/INTESTINE | URINARY BLADDER | SKIN |
|---|---|---|---|---|---|---|---|---|
| EFFECT | Miosis (constrict/ pinpoint), increased lacrimation | Increased secretions | Increased secretions | Decreased heart rate | Bronchoconstriction bronchorrhea (increased bronchial secretions) | Increased GI secretions, increased vomiting, increased intestinal motility (diarrhea) | Increased urination | Increased sweating |
| RECEPTOR | M3 | M3 | M3 | M2 | M3 | M3 | M3 | M3 |

What are the physiologic effects of stimulating the PNS in the lungs?

## What adverse effects are associated with AChE inhibitors? (LO#4)

Systemic stimulation of acetylcholine receptors can exaggerate the body's parasympathetic or cholinergic response. Both the reversible and irreversible cholinergics exert these effects. Depending on the dose used these effects can turn into adverse effects.

The commonly used toxidrome used to identify cholinergic toxicity is DUMBBELLS: The toxidrome lists the muscarinic symptoms can be remembered with the mnemonic **DUMBBELLSS:** **D**efecation, **U**rination, **M**iosis, **B**radycardia, **B**ronchospasm, **E**mesis, **L**acrimation, **L**ethargy, and **S**alivation

Mnemonic

Adverse effects of cholinomimetics = **DUMBBELLSS**: **D**efecation, **U**rination (incontinence), **M**iosis, **B**ronchoconstriction, **B**radycardia (with high doses), **E**mesis, **L**acrimation, **L**ethargy, **S**alivation and **S**weating

In addition, activation of nicotinic effects can include muscle fasciculations, weakness, and paralysis.

Indirect-acting cholinomimetics do not cause vasodilation, since the vasculature is not normally innervated by the PNS. Therefore, these medications may lead to bradycardia with reflex vasoconstriction to maintain a steady cardiac output.

The adverse effects are more pronounced with the irreversible AChE inhibitors (Organophosphates).

How do indirect-acting cholinomimetics treat myasthenia gravis?

## What Is Cholinesterase Inhibitor Poisoning? (LO#5 and 6)

There are many choli ironment, especially around farms where they are used in pesticides. Farmers and their families, who are exposed to organophosphate pesticides like parathion and malathion, are at risk for cholinesterase inhibitor poisoning. Organophosphates such as sarin and soman are used for chemical warfare. For instance, sarin is a neurotoxic gas used as a chemical agent to poison populations during recent wars. These agents have very short aging times. For instance, with soman, aging occurs in 10 minutes.

### Signs and Symptoms of Cholinesterase Inhibitor Toxicity

The signs and symptoms of cholinesterase inhibitor poisoning can be deduced from the adverse effects we described previously. These include:

- Drooling due to increased secretion from the salivary glands
- Dyspnea due to bronchoconstriction, increased bronchial secretions, and bronchospasm. Bronchorrhea due to excessive secretions also occurs
- Excessive sweating (diaphoresis)
- Miosis due to constriction of the pupillary sphincter muscles
- Muscle fasciculations (twitching) due to constant acetylcholine nicotinic activation
- Defecation due to excessive GI motility
- Urinary incontinence due to relaxation of the urinary sphincter and bladder contraction

## How can we treat Cholinesterase Inhibitor Toxicity? (LO#6)

Poisoning with orga nale behind the treatment is one that blocks the excessive ACh from binding to (and activating) ACh receptors, at least until the drug or agent has been excreted from the body.

**Atropine**, a competitive antagonist of muscarinic ACh receptors, is used to reverse acute cholinesterase inhibitor toxicity. They are more effective in toxicity caused from reversible AChE inhibitors. Atropine cannot restore the function of the AChE enzyme and hence cannot be the only agent used for cholinergic toxicity mediated by irreversible AChE inhibitors. In such cases, atropine is used along with pralidoxime (2-PAM).

The drug **pralidoxime (2-PAM)** has a separate role for treating poisoning. Pralidoxime, an oxime is a reactivator. Pralidoxime binds to a separate site on AChE (not the one to which ACh binds) and can prevent this aging from occurring. In fact, pralidoxime can displace the organophosphate from its binding site and regenerate normal cholinesterase function. If given before aging has occurred, strong nucleophiles like PAM are able to break the phosphorus-enzyme bond and hence are used as "cholinesterase regenerator" drugs for organophosphate poisoning.

Once aging has occurred, the enzyme-inhibitor complex is even more stable and is more difficult to break, even with oxime regenerator compounds.

What drugs are used to treat organophosphate poisoning?

**CASE CONNECTION**

BACK TO INTRODUCTION ↑

Thinking back to JK, what the rationale for administration of 2-PAM and atropine to JK?

JK's signs and symptoms point to cholinergic poisoning. JK has been exposed to parathion, an organophosphate used as an insecticide. JK is showing all the signs of the DUMBBELLS toxidrome for cholinergic poisoning. He is defecating, urinating, miosis (pinpoint pupils), bradycardic (HR of 45 bpm), lacrimation, and salivating. He is also in considerable respiratory distress. Before aging sets in for the organophosphate, the physician must administer the antidote for organophosphate poisoning. The rationale behind ordering atropine and 2-PAM (pralidoxime) for JK is to counteract the effects of organophosphate poisoning. Atropine works by blocking the action of acetylcholine at muscarinic receptors, which helps to reduce the excessive salivation, sweating, bronchoconstriction, and other cholinergic symptoms that JK is experiencing. This helps to alleviate respiratory distress and other symptoms caused by the overstimulation of the parasympathetic nervous system. Pralidoxime (2-PAM) reactivates acetylcholinesterase, the enzyme that is inhibited by organophosphates like parathion. By reactivating this enzyme, 2-PAM helps to break down acetylcholine, reducing its accumulation and thereby alleviating the symptoms of poisoning. This combination of treatments addresses both the immediate symptoms and the underlying cause of the poisoning, providing a comprehensive approach to managing JK's cond

## Summary

- Cholinomimetic agents mimic acetylcholine and activate the parasympathetic nervous system by binding to muscarinic receptors.
- Direct-acting cholinomimetics bind directly to acetylcholine receptors, while indirect-acting agents inhibit acetylcholinesterase, increasing acetylcholine in the synaptic cleft.
- Reversible indirect-acting cholinomimetics like neostigmine and physostigmine are used to treat conditions such as glaucoma, myasthenia gravis, and atropine overdose.
- Irreversible cholinesterase inhibitors, such as organophosphates, are used in insecticides and chemical warfare, and can cause severe toxicity.
- Cholinesterase inhibitor poisoning is treated with atropine and pralidoxime, which block acetylcholine receptors and regenerate cholinesterase function, respectively.
- Adverse effects of cholinomimetics include symptoms like diarrhea, urination, miosis, bronchoconstriction, bradycardia, emesis, lacrimation, and salivation/sweating, summarized by the mnemonic DUMBBELSS.

## Review Questions

**1. Which of the following can irreversibly phosphorylate the AChE enzyme?**

A. ○ Neostigmine
B. ● Malathion
C. ○ Physostigmine
D. ○ Pyridostigmine

**Hide Explanation**

The correct answer is Malathion (B). Malathion is a irreversible AChE inhibitor which can cause phosphorylation of the AChE enzyme. Phosphorylation of the AChE enzyme leads to reversible inhibition of the enzyme. Neostigmine (A), Physiostigmine (C), and Pyridostigmine (D) are reversible AChE inhibitors that carbamolyate AChE. This is a reversible process that can restore enzyme function.

**2. Which of the following cholinergic agents can cross the blood-brain barrier?**

A. ○ Neostigmine
B. ○ Pyridostigmine
C. ● Physiostigmine

**Hide Explanation**

The correct answer is amine which crosses the BBB. Neostigmine (A) and pyridostigmine (B) are 3° amines and cannot cross the BBB.

**3. Which of the following is an example of an indirect-reversible acting AChE inhibitor?**

A. ○ Bethanechol
B. ○ Carbachol
C. ● Neostigmine
D. ○ Pilocarpine

**Hide Explanation**

The correct answer is Neostigmine (D). Neostigmine is a direct-acting AChE inhibitor. Bethanechol (A), carbachol (B), and pilocarpine (D) are also direct-acting cholinomimetic agents, but they are not used to test for asthma.

**4. What is the primary function of pralidoxime (2-PAM) in organophosphate poisoning?**

A. ○ Inhibits ACh release
B. ● Regenerates AChE activity
C. ○ Binds to ACh receptor
D. ○ Facilitates AChE aging
E. ○ Blocks ACh receptors

**Hide Explanation**

The correct answer is 'Regenerates AChE function' (B). Pralidoxime works by displacing the organophosphate from its binding site and regenerating normal cholinesterase function. 'Inhibits ACh release' (A) is incorrect because pralidoxime does not affect ACh release. 'Binds to ACh' (C) is incorrect as pralidoxime binds to a separate site on AChE, not ACh. 'Facilitates AChE aging' (D) is partially correct but not the primary function; it prevents aging by breaking the phosphorus-enzyme bond. 'Blocks ACh receptors' (E) is incorrect as pralidoxime does not block ACh receptors.

**5. A case of physostigmine overdose was diagnosed. Choose the reversal agent from the following list.**

A. ● Atropine
B. ○ Donepezil
C. ○ Malathion
D. ○ Pilocarpine

**Hide Explanation**

The correct answer is Atropine (A). Atropine is a anti-choligernic agent or a cholinergic inhibitor. It blocks ACh from binding to the cholinergic receptors and hence can help in reversing physostigmine overdose. Donepezil (B) is an indirect-acting reversible AChE inhibitor and cannot be helpful in reversing ACh toxicity. (C) Malathion is an irreversible-acting AChE inhibitor that can exacerbate ACh-mediated toxicity and hence cannot be of use here. It is an insecticide and should be used for clinical purposes. Pilocarpine (D) is a direct-acting cholinergic agent.

**6. If an overdose from Neostigmine occurs, which of the following effects will be observed?**

A. ○ Mydriasis
B. ● Miosis
C. ○ Bronchodilation
D. ○ Urinary retention

**Hide Explanation**

The correct answer is t-AChE inhibitor, it will increase Ach at the erated, hence Miosis in the eyes could occur. Mydriasis (B) is an effect exhibited by anti-cholinergics. Cholinergic, such as neostigmine, will cause bronchoconstriction; hence bronchodilation (C) is incorrect. A systemic overdose of cholinergic will result in urination; hence urinary retention (D) is an incorrect answer.

**How would you rate this Brick?**

### Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast the mechanism of action of direct- and indirect- acting AChE inhibitors.~~
02 ☑ ~~Describe the pharmacokinetic properties of various direct- and indirect- acting AChE inhibitors.~~
03 ☑ ~~Recognize the clinical signs and symptoms of cholinergic toxidromes and link them to cholinergic receptor activation.~~
04 ☑ ~~Explain how cholinesterase reactivators (oximes) reverse organophosphate poisoning.~~
05 ☑ ~~Provide the rationale for the use of atropine as an antidote for cholinergic poisoning.~~
06 ☑ ~~Discuss in chemical~~

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Objective Confidence: 6/6